Compositional Bias in Single-Cell RNA-seq Differential Expression

Are lowly or highly expressed passenger genes more affected?

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The question

In single-cell RNA-seq differential expression (DE) testing, the data are *compositional*: because of library-size (total-count) normalization, each cell is effectively a vector of relative abundances. If you artificially change the expression of certain ("driver") genes, the apparent expression of other ("passenger") genes can shift too — even though their absolute expression did not change.

Which passenger genes are more affected: lowly or highly expressed ones?

Short answer

The spurious **fold-change is essentially the same size for every passenger gene**, but the spurious change is far more likely to be **detected** (called significant) in **highly expressed genes**. So in practice it is the highly expressed genes that *look* differentially expressed.

Why the bias is a constant offset in log space

Library-size normalization turns each cell into fractions,

$$p_g = \frac{c_g}{T}, \qquad T = \sum_g c_g,$$

where c_g is the raw count of gene g and T is the cell's total count.

Suppose some driver genes gain ΔT extra counts, so the new total is $T' = T + \Delta T$. A passenger gene's *absolute* counts c_g are unchanged, but its normalized value becomes

$$p'_g = \frac{c_g}{T'} = p_g \cdot \frac{T}{T'}.$$

The factor T/T' is **the same for every passenger gene**. In terms of log-fold-change,

$$\log \frac{p'_g}{p_g} = \log \frac{T}{T'} = \text{constant, independent of } g.$$

So the *magnitude* of the artifactual logFC does **not** depend on whether a passenger gene is low- or high-expressed: every unchanged gene is shifted down by the same amount.

Why detectability depends on expression level

Whether that constant shift crosses the significance threshold depends on the gene's **noise**, not just its effect size. The test statistic is roughly

$$z \approx \frac{\text{shift}}{\text{SE}}.$$

Counts are sampled (Poisson / negative-binomial), so:

- **Highly expressed genes** have low *relative* sampling noise (small coefficient of variation, few dropouts) $\Rightarrow$ small SE $\Rightarrow$ the constant compositional shift produces a large, significant statistic. **These light up as false positives.**
- **Lowly expressed genes** have high relative noise and many zeros $\Rightarrow$ large SE $\Rightarrow$ the same shift is buried in noise and rarely reaches significance.

Conclusion: highly expressed passenger genes are the ones more affected in terms of spuriously appearing DE.

A second-order effect on the driver side

Changing a *highly* expressed gene moves T a lot (large ΔT, hence a large T/T' shift on everyone), whereas perturbing a *lowly* expressed gene barely changes the total and causes almost no compositional artifact. For the passenger genes the question is about, however, it is still the highly expressed ones that get falsely called.

Practical implications

- A few strongly induced, abundant genes can create a "smear" of apparent down-regulation across many other genes — concentrated among high-mean genes with a consistent sign. That pattern is a signature of a compositional artifact rather than real biology.
- Robust size-factor methods resist this better than plain total-count normalization, because they estimate the normalization from the bulk / median of genes rather than letting a few dominant genes set T: DESeq2 (median-of-ratios), edgeR (TMM), scran (pooled size factors).
- Sanity check your hits: if they are dominated by high-mean genes that all move in the same direction, suspect compositionality.

Relevance to this project

The pipeline normalizes to a fixed `target_sum` (200000) followed by `log1p` — i.e. total-count normalization — so this caveat applies directly. The glmGamPoi path models raw counts with size factors and is more robust; the Welch-on-`log1p` path is more exposed to the artifact. This is a useful lens for the `CompareWelchVsGlmGamPoi` comparison.